ODE-FFN: Progress Report 1

Independent Study — Deep Learning

Zhizhe Zhang
NYU Shanghai

20 October 2026

1 The project, in short

This project asks whether a feed-forward block built from a learned dynamical operator is better suited to flow matching, which learns a velocity field transporting noise to data [1]. The intended application is to replace the FFN in a JiT image generator [2]. Before returning to images, we use small, unconditional synthetic problems to test what the original ODELayer actually contributes.

The main progress is a controlled comparison of constraints on P, gating, Taylor order, and training budget, with SiLU and SwiGLU [3] baselines. The latest stage completed **30 training trajectories and 54 checkpoint diagnostics**; earlier runs supply the spectral-cap comparison. The main result is mixed: ODE $K = 2$ beats SwiGLU at equal updates on all three targets, but only on Moons after matching a compute proxy. A smaller internal Jacobian and a more accurate exponential approximation do not consistently improve generation.

2 Checking the layer before extending it

The professor's original layer uses a learned matrix P and a scalar gate $s(h)$:

$$F_K(h) = T_K(s(h)P)h, \qquad T_K(A) = \sum_{k=0}^{K} \frac{A^k}{k!}, \qquad s(h) = 0.2 + 0.8\ \mathrm{sigmoid}(g(h)/10). \tag{1}$$

Here g is a two-layer SiLU network. Each layer computes s once and shares P across Taylor terms. The FFN contains two such layers with SiLU between them; at $K = 1$, one layer is simply $h + s(h)Ph$. For fixed s, $\exp(sP)h$ solves $dy/du = Py$ over duration s. This hidden duration is distinct from generation time t.

What needs control? The exponential series converges for every finite P; the issue is finite-order error and amplification. Differentiating the layer gives

$$J_{F_K} = T_K(sP) + [P\,T_{K-1}(sP)h]\,\nabla_h s^\top. \tag{2}$$

A bound on P therefore does not by itself control the input gate or the full velocity network. This standard identity motivates measuring all three quantities rather than treating P as the Jacobian of the generated flow.

Data and controls. We use an anisotropic Gaussian with mean $(1, -1)$ and covariance $\mathrm{diag}(0.09, 4)$; eight equally weighted Gaussian modes on a radius-3 ring, each with covariance $0.04I$; and centered two moons with noise standard deviation 0.08. These are specified toy distributions, not an exact paper benchmark reproduction. Each has 100,000/10,000/10,000 train/validation/test points. Training predicts $x_1 - x_0$ from $x_t = (1 - t)x_0 + tx_1$ and t, using independent Gaussian source noise, uniform t, and no labels. All runs use seed 0, width 64, three residual blocks, batch 512, AdamW at 3×10^{-4}, and a default endpoint of 5,000 updates.

Evaluation. The primary metric is sliced Wasserstein distance SW_2 (lower is better): the root mean squared difference of sorted projections, using 128 fixed directions and 2,048 fixed generated/target points. Sampling uses 32 Heun steps. Gaussian and Ring8 also have analytical velocity references; Moons does not. All results are single-seed observations.

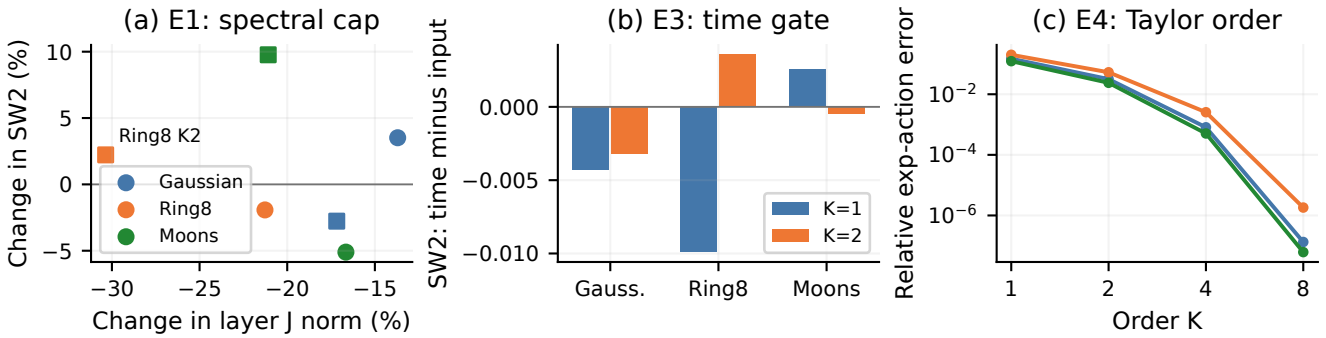


Figure 1: (a) Change from free to capped P; circles: $K = 1$, squares: $K = 2$. (b) Time-gate minus input-gate SW_2; negative is better. (c) Relative output discrepancy against $\exp(sP)h$; dataset colors follow (a). Orders are separately trained. All panels use seed 0 and 5,000 updates.

Model / updates	Gaussian	Ring8	Moons
SiLU / 5,000	0.0868	0.1631	0.0448
SwiGLU / 5,000	0.0746	0.2080	0.0302
ODE K2 / 5,000	0.0674	0.1848	0.0271
SwiGLU / 6,572	0.0658	0.1508	0.0624

Table 1: Fixed-endpoint SW_2 at Heun32; seed 0. The final row matches ODE $K = 2$'s forward-linear-MAC budget. These are individual runs, not means over training seeds.

3 E1–E2: do constraints and extra computation help?

E1, spectral cap. Purpose: test whether reducing internal amplification improves samples. We compared free P with $P_{\text{eff}} = P \min(1, 1/\|P\|_2)$ at $K = 1, 2$. The cap lowers the full-layer Jacobian norm in all six comparisons, but improves SW_2 in only three (Fig. 1a). On Ring8 at $K = 2$, the Jacobian norm falls from 3.014 to 2.099, while SW_2 worsens from 0.1848 to 0.1889. On Gaussian, the same intervention improves SW_2 from 0.0674 to 0.0656. Thus smaller hidden-layer gain is not a reliable performance target. The norms average 16 observed inputs per layer and six layers; they are not global bounds.

E2, baseline and budget. Purpose: check whether an ODE advantage survives a competitive baseline and a computation control. Parameter counts are close: 82,056 for ODE, 81,858 for SiLU, and 81,792 for SwiGLU. At 5,000 updates, ODE $K = 2$ beats SwiGLU on all three targets, but SiLU is better on Ring8 (Table 1). ODE $K = 2$ costs 105,152 forward linear multiply–accumulates (MACs) per example versus 80,000 for SwiGLU. At 6,572 updates, SwiGLU matches this cumulative MAC budget and has lower SW_2 on Gaussian and Ring8; ODE remains better on Moons. This is not a uniform advantage. MACs exclude backward, nonlinear, and optimizer costs, so the comparison is not a GPU-time match.

4 E3–E5: which part of ODELayer matters?

E3, input versus time gating. Purpose: test the input-dependent step size. Feeding the same-width time embedding to g preserves parameter count and initial weights but removes the second term of Eq. (2) at fixed t. The outcome is not uniformly better (Fig. 1b). On Moons at $K = 2$, the full Jacobian norm rises from 2.334 to 2.509 despite the zero gate-derivative term; SW_2 changes from 0.02708 to 0.02663. Retrained P and gate values can compensate. This changes gate inputs and optimization as well as the derivative term, so it is not a pure derivative intervention.

E4, Taylor order. Purpose: test whether a more accurate hidden ODE solve helps. We trained input-gated models with $K = 1, 2, 4, 8$. Going from $K = 4$ to $K = 8$ adds **63.7%** forward linear MACs, yet changes each target's SW_2 by less than **0.2%**. On Ring8, exponential-action discrepancy falls from 2.55×10^{-3} to 1.82×10^{-6} (Fig. 1c), while SW_2 only changes from 0.17518 to 0.17483.

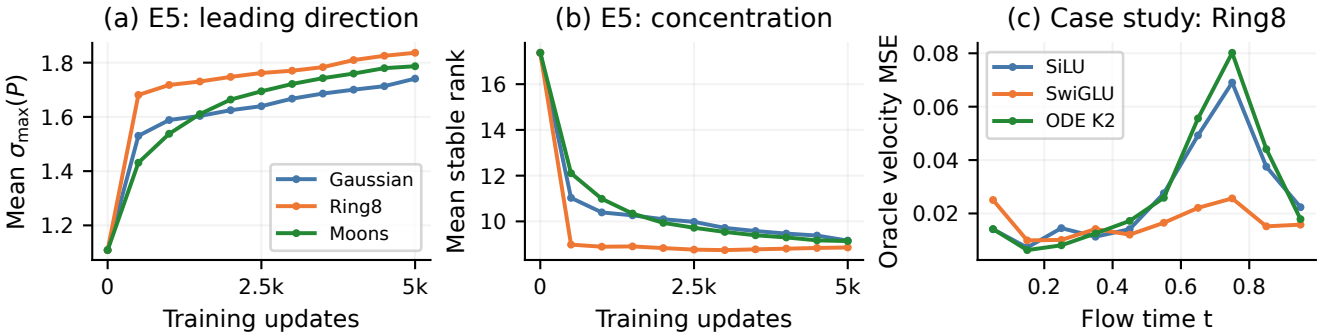


Figure 2: (a–b) Original $K = 2$: six-layer means of the largest singular value and stable rank. E5's endpoint ranges cover all four orders; layers are not independent replicates. (c) Ring8 oracle velocity MSE on held-out interpolated points, not generated trajectories. All panels use seed 0.

Gaussian $K = 1$ also generates better than $K = 2$ despite being less accurate against the exponential. Higher order can help—Ring8 improves from $K = 1$ to $K = 4$—but exponential accuracy alone does not explain the results.

E5, what P learns. Purpose: inspect training dynamics rather than only the final score. Across the three targets and four orders, the six-layer mean largest singular value rises from about 1.109 to 1.715–1.862, while stable rank $\|P\|_F^2/\|P\|_2^2$ falls from 17.37 to 8.31–9.40 (Fig. 2a–b shows $K = 2$). The $K = 4$ and $K = 8$ trajectories nearly overlap. Energy concentrates relative to the strongest direction, without implying a loss of algebraic rank. Flattening the spectrum could therefore remove useful direction selection.

5 Case study: fitting error is not sample quality

On Ring8, SwiGLU has lower aggregate oracle velocity MSE than ODE $K = 2$ (0.01681 versus 0.02871), yet worse generation at equal updates. Relative to SiLU, it fits $t \in [0.7, 0.8)$ better (0.0256 versus 0.0690), but $t \in [0, 0.1)$ worse (0.0250 versus 0.0141; Fig. 2c). Error timing is a candidate explanation, not a demonstrated cause. Increasing Heun steps from 32 to 256 preserves the principal Ring8 ranking, weakening a solver-only explanation.

Endpoint sensitivity also limits the budget comparison: Ring8 SiLU changes from 0.1631 at update 5,000 to 0.2258 at 5,024. We retain preselected endpoints, not best test scores. One seed and repeated test-set inspection limit the conclusions. These toy models have no attention and predict velocity, unlike JiT's image prediction; image benefits remain untested.

6 Hypotheses for the next two weeks

1. **Hypothesis:** $K = 4/K = 8$ similarity mainly reflects sufficient evaluation accuracy. **Test:** freeze checkpoints and vary only the evaluation order or use $\exp(sP)$. **Expected:** $K = 4$ and $K = 8$ substitutions have little effect on velocity and samples. Large changes would instead suggest compensation during training.
2. **Hypothesis:** operator geometry matters beyond eigenvalue magnitude. **Test:** vary non-normality at matched spectral radius, then repeat after matching spectral norm; measure polynomial and exponential gain on common probes. **Expected:** some gain differences remain after norm matching; disappearance would favor scale as the simpler explanation.
3. **Hypothesis:** input gating adds more than a learned scalar scale. **Test, if this direction is selected:** compare the original gate with fixed $s = 0.6$ and a learned scalar per layer, retaining seed 0, both baselines, and compute accounting. **Expected:** an input-gate benefit survives a validation-based endpoint rule fixed before training. These are proposed tests, not completed results.

References

[1] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow Matching for Generative Modeling. *arXiv:2210.02747*, 2022; revised 2023. https://arxiv.org/abs/2210.02747.

[2] T. Li, K. He. Back to Basics: Let Denoising Generative Models Denoise. *arXiv:2511.13720*, 2025; revised 2026. https://arxiv.org/abs/2511.13720.

[3] N. Shazeer. GLU Variants Improve Transformer. *arXiv:2002.05202*, 2020. https://arxiv.org/abs/2002.05202.